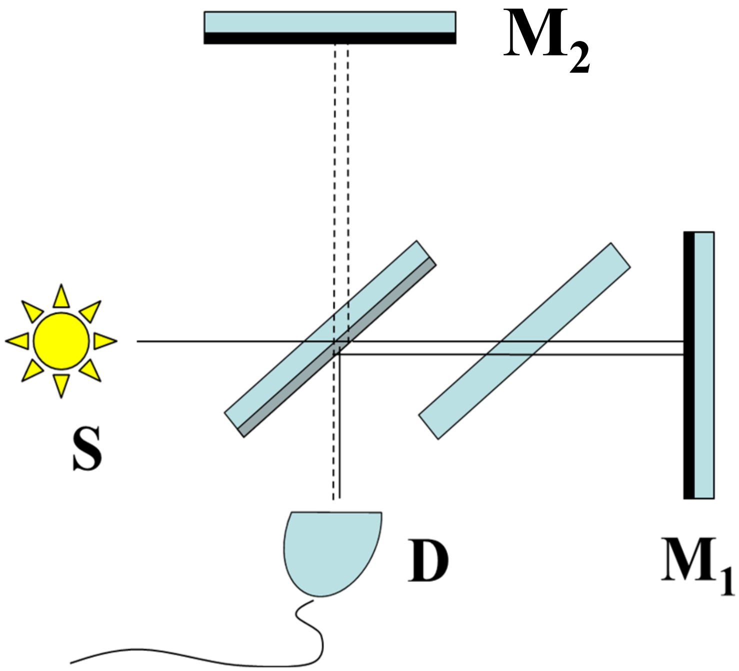
# Ph134, Lecture 9

- **The coherence function $g^{(1)}(\tau)$**
- **Properties of the coherence function $g^{(1)}(\tau)$**
- **Two-wavelength input: sodium doublet**
- **Sources with continuous wavelength/frequency distributions**

# Input and output intensities:




$E(t)$
$=$ scalar electric field of light leaving S at time $t$

$t_1 =$ time from S to D via path 1
$t_2 =$ time from S to D via path 2

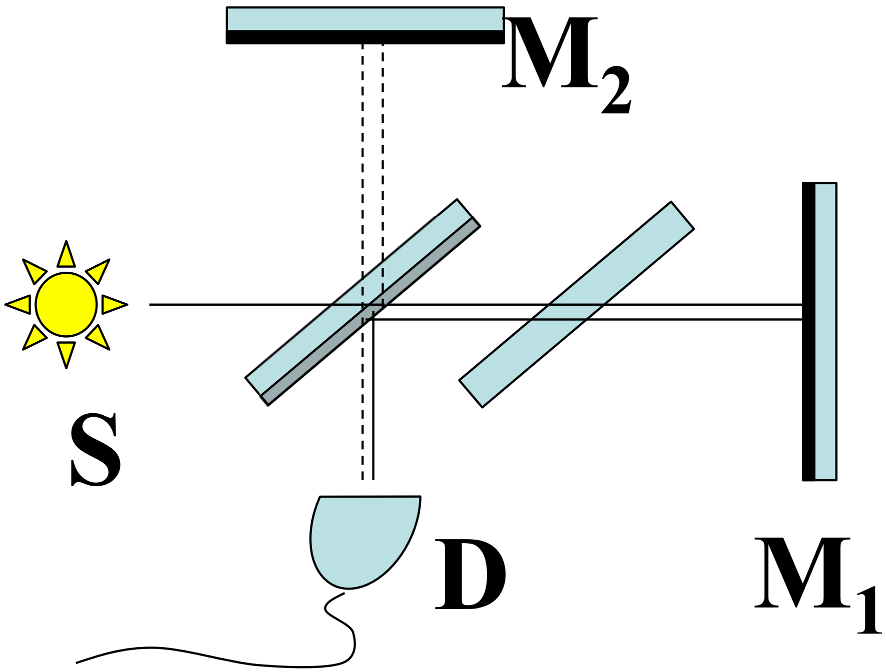
$$\tau = t_2 - t_1$$

$$I_{OUT} = \frac{1}{\mu_0 c}\left[\left\langle E^{(+)}(t)E^{(-)}(t)\right\rangle - \mathrm{Re}\left\langle E^{(+)}(t+\tau)E^{(-)}(t)\right\rangle\right]$$

$$I_{OUT} = \frac{1}{2} I_{IN}\left[1 - \mathrm{Re}\; g^{(1)}(\tau)\right]$$

$$where\; g^{(1)}(\tau) \equiv \frac{\left\langle E^{(+)}(t+\tau)E^{(-)}(t)\right\rangle}{\left\langle E^{(+)}(t)E^{(-)}(t)\right\rangle}$$

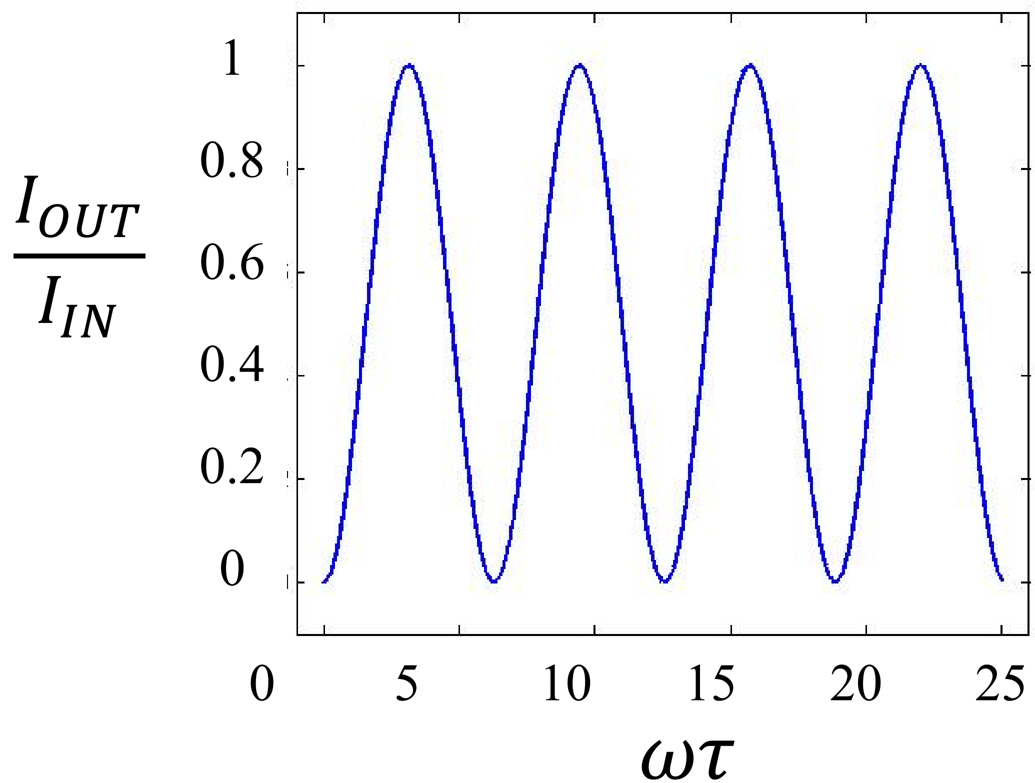
# Monochromatic source:




$$E^{(+)}(t) = E_0^{(+)} e^{-i\omega t}$$

$$g^{(1)}(\tau) = e^{-i\omega\tau}$$

$$I_{OUT} = \frac{I_{IN}}{2}(1 - \cos\omega\tau)$$

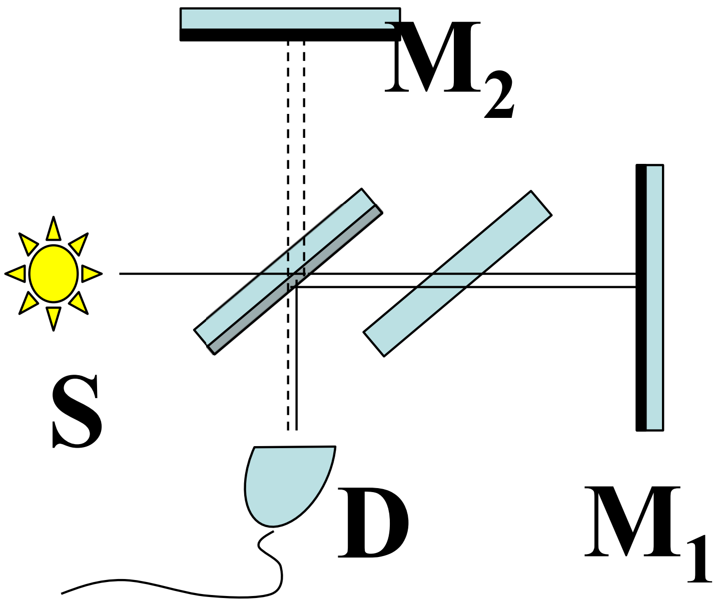



*Complete "interference fringe" when $\tau$ changes by $\frac{2\pi}{\omega}$.*

$$\tau = \frac{path\ length\ 2\ - path\ length\ 1}{c}$$

*Complete "interference fringe" when path length difference changes by $\lambda$.*

# Two-wavelength source (*e.g.*, Na doublet):

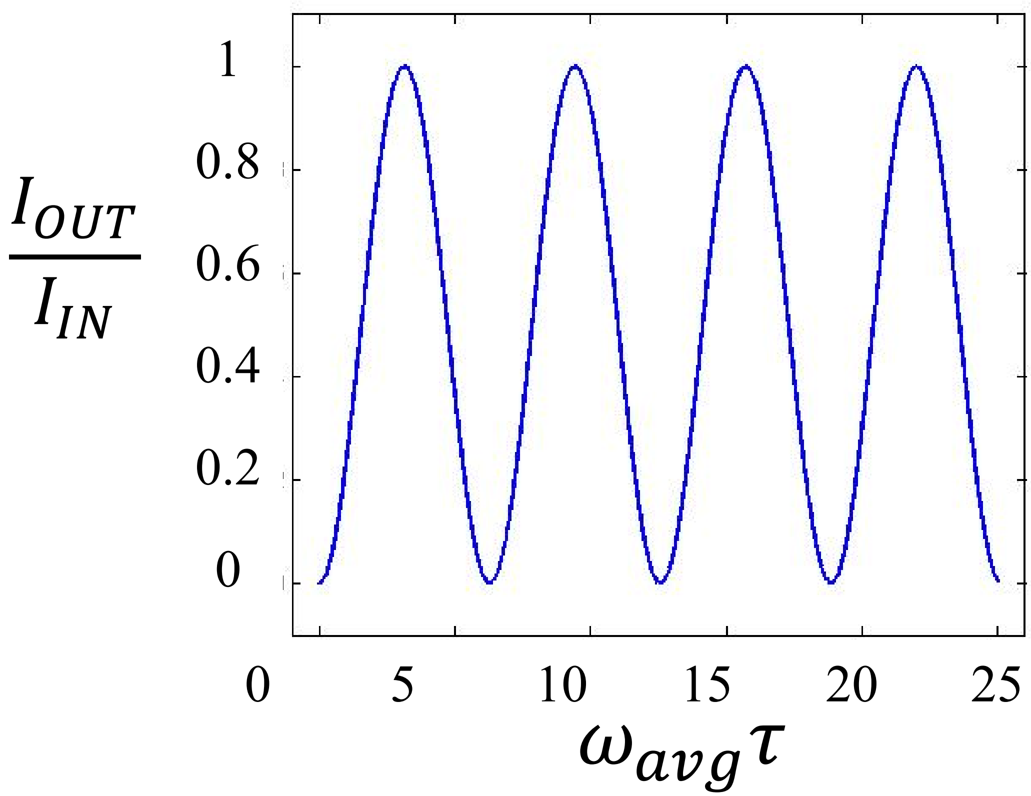


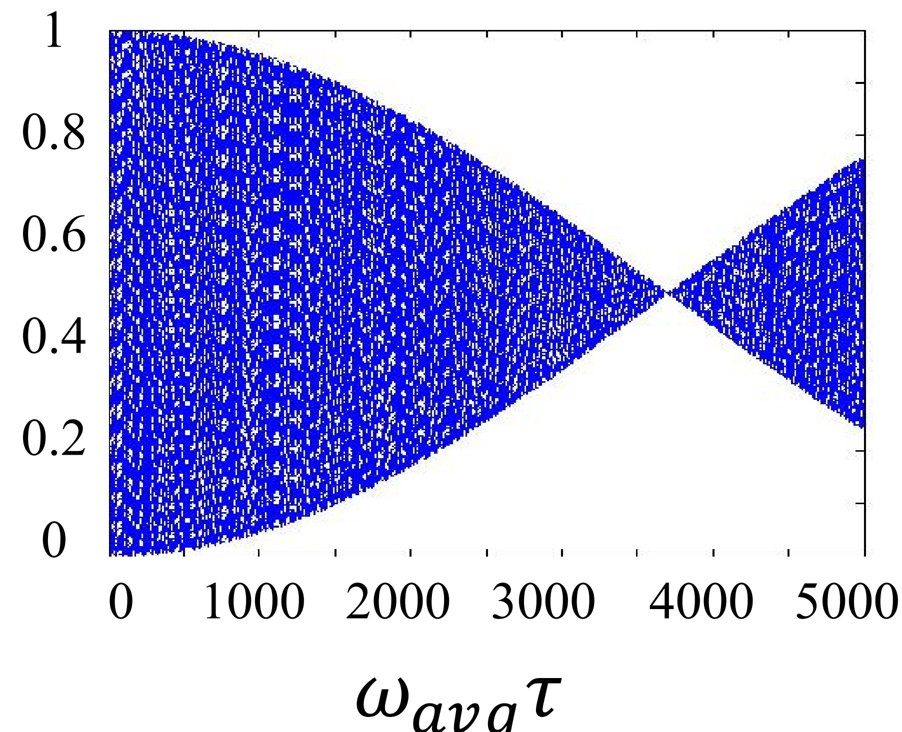

$$E^{(+)}(t) = E_0^{(+)} e^{-i\omega_1 t} + E_0^{(+)} e^{-i\omega_2 t}$$

$$\omega_1 - \omega_2 \gg \frac{2\pi}{T_{det}} \Rightarrow$$

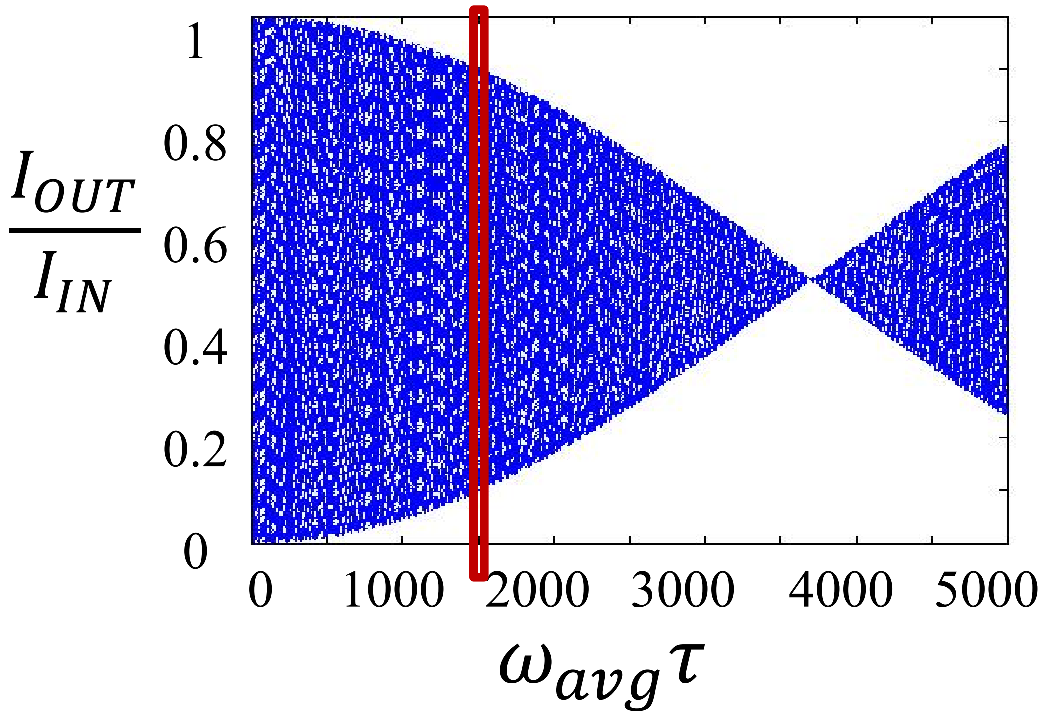
$$g^{(1)}(\tau) = \frac{e^{-i\omega_1\tau} + e^{-i\omega_2\tau}}{2}$$

$$I_{OUT} = \frac{1}{2} I_{IN} \left(1 - \frac{\cos\omega_1\tau + \cos\omega_2\tau}{2}\right)$$



# Source coherence function at time delay τ related to <u>fringe visibility</u>:




$$V(\tau) \equiv \frac{I_{OUT,max} - I_{OUT,min}}{I_{OUT,max} + I_{OUT,min}}$$

*Max, min are over a small range of delays near τ. This concept assumes fringe amplitude varies insignificantly over one fringe period.*

$$\boxed{V(\tau) = \left|g^{(1)}(\tau)\right|}$$

*For a monochromatic source,* $V(\tau) = \left|g^{(1)}(\tau)\right| = 1.$

# Multi-wavelength source?

$$E^{(+)}(t) = \sum_j E_j^{(+)} e^{-i\omega_j t}$$

$$\omega_j - \omega_k \gg \frac{2\pi}{T_{det}}$$

$$\Longrightarrow \quad I_{IN} = \sum_j \frac{2\left|E_j^{(+)}\right|^2}{\mu_0 c} = \sum_j I_j$$

*So it makes sense to talk about the input power as having a certain fraction $p(\omega_j)$ at each frequency. Then:*

$$g^{(1)}(\tau) = \sum_j p(\omega_j) e^{-i\omega_j \tau}$$

$$\mathrm{Re}\, g^{(1)}(\tau) = \sum_j p(\omega_j) \cos \omega_j \tau$$

***Fringe visibility depends on the individual monochromatic interference patterns being in sync with each other or not.***

# Source with continuous wavelength distribution?

$$E^{(+)}(t) = \int_0^\infty \mathcal{E}^{(+)}(\omega) e^{-i\omega t} d\omega$$

$$I_{IN} = \int_0^\infty \tilde{I}(\omega) d\omega$$

angular frequency bandwidth (width of the distribution in angular frequency space) $\gg \dfrac{2\pi}{T_{det}}$

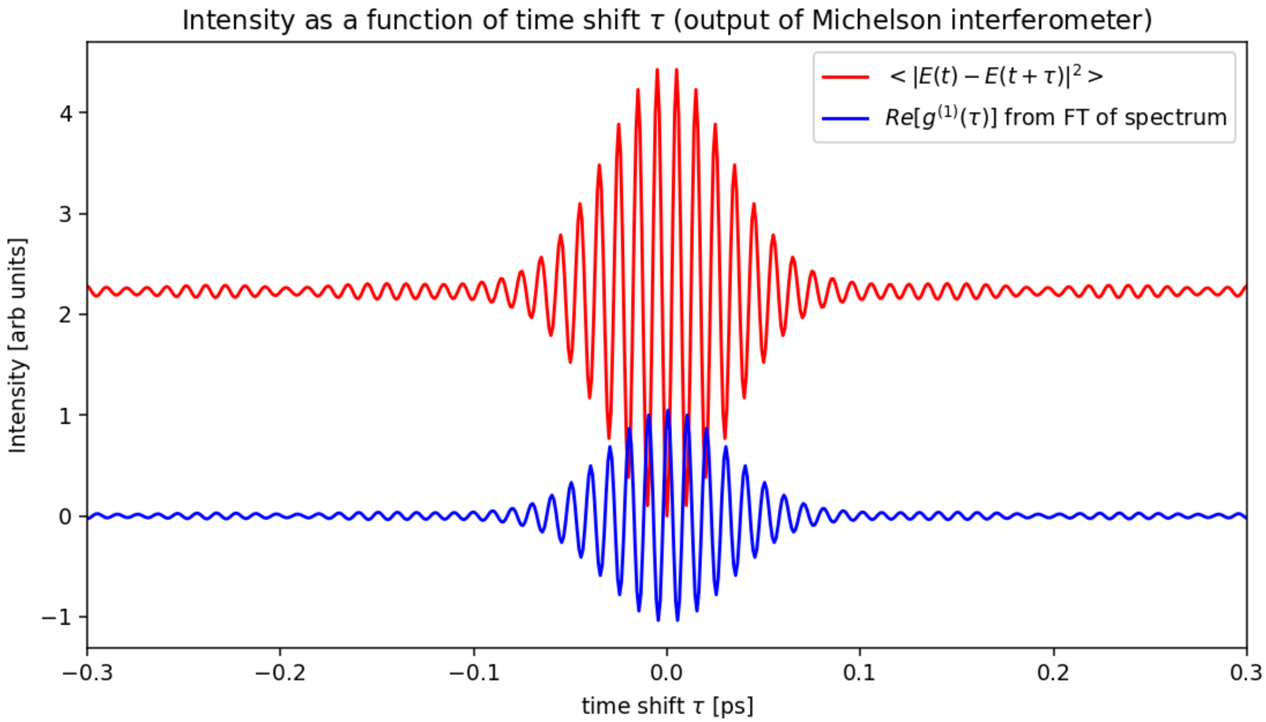
$\Rightarrow$ *So it makes sense to talk about the input power as having a certain fraction $s(\omega)d\omega$ in each small frequency interval. Then:*

$$g^{(1)}(\tau) = \int_0^\infty s(\omega) e^{-i\omega\tau} d\omega$$

$$\mathrm{Re}\, g^{(1)}(\tau) = \int_0^\infty s(\omega) \cos\omega\tau \, d\omega$$

***Fringe visibility depends on the individual monochromatic interference patterns being in sync with each other or not. Once patterns get out of sync, no "recovery" at larger τ. Limited range around τ=0 where coherence is present / fringes observed.***

# Coherence time and coherence length:




$$\tau_{coh} \equiv \int_{-\infty}^{\infty} \left(\mathrm{Re}\left[g^{(1)}(\tau)\right]\right)^2 d\tau \qquad \ell_{coh} = c\tau_{coh}$$

$$(\tau_{coh})(\text{source bandwidth}) \sim 1$$